Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

- Quantity of heat for (a) = m âĭħ c âĭħ ÎĶT = 1000 g âĭħ 1 cal/gÂ°C âĭħ (60Â°C - 30Â°C) = 3000 cal or 30 Kcal.
- Quantity of heat for (b) = m âĭħ c âĭħ ÎĶT = 1000 g âĭħ .215 cal/gÂ°C âĭħ (60Â°C - 30Â°C) = 645 cal or 6.45 Kcal.

So, the quantity of heat required is 30 Kcal for water and 6.45 Kcal for aluminum.

</think>

<answer>

G. 30 Kcal, 6.45 Kcal

</answer>